

Neuromechanical Signatures of Human-Hip Exoskeleton Interaction: Protocol & Preliminary Observations

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The Question

Exoskeletons are tuned to save energy. Is energy what the wearer is optimizing?

Human-in-the-loop optimization usually minimizes metabolic cost [1, 2], while user preference is emerging as an alternative objective [3]. We combine a custom questionnaire with multimodal physiological recordings to link how wearers **evaluate** the interaction to signatures in their **body** (EMG) and **brain** (EEG) activity.

Protocol & Setup

Two days, the same eight blocks

Day 1 (GRF, IMU, EMG, Calorimetry, Questionnaire)

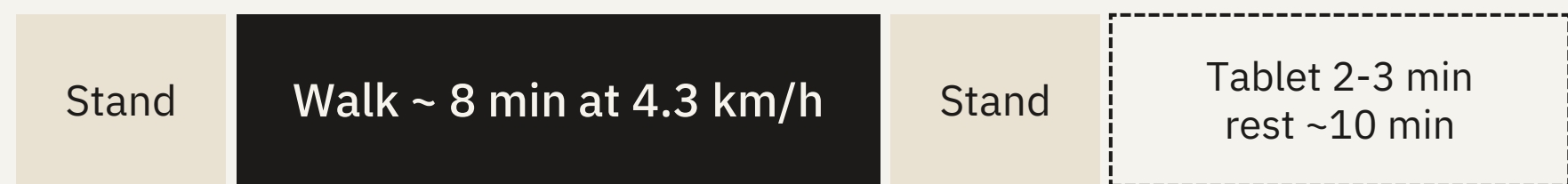
Day 2 (GRF, IMU, EMG, EEG, Questionnaire)



no exo + assistive modes zero torque - resistive modes

Exo modes in randomized order (example shown); the participant never learns which mode is running

One block



Device. Dnsys X1 hip exoskeleton, commercial, with a closed controller.

Participants. 4 healthy adults, 22-26 years old (planned for 20)

EEG. 64-channel actively amplified wet Ag/AgCl electrodes, 500 Hz (Brain Product GmbH, Germany)

EMG. Trigno Avanti sensors on lower limb muscles (Soleus, Gastrocnemius medial, Tibialis anterior, Vastus medialis, Rectus femoris, Biceps femoris, Gluteus maximus); 8 channels of Trigno Dou sensors on Neck muscles (Trapezius, SCM), 2048 Hz (Delsys Inc., USA)

IMU. MTw Awinda sensors x7 (lower limbs and Pelvis), 100 Hz (Xsens, Netherland)

GRF. Instrumented treadmill (dual force plates)

Calorimetry. COSMED K5, Wearable, backpack-style portable metabolic system (COSMED, Italy)

Sync. All streams (except K5) and markers on Lab Streaming Layer [6], one XDF per block, aligned on per-sample LSL timestamps.

EMG sensor
IMU sensor



What the pilot shows?

The device changes the energetics. Resistive modes raised net metabolic cost by about 10 to 25 % relative to zero torque; high assistance lowered it by about 5 % on first use.

The strongest help did not feel like help. Three of four participants rated high assistance as hindering, although it was the mode they felt acting most clearly.

Pleasantness follows how walking felt, not energy. Within participants it tracked naturalness ($r_{rm} = +0.90$) and restriction (-0.78); net metabolic cost barely did (-0.26 , 95 % CI -0.57 to $+0.12$) [6].

People weigh the terms differently. Predictability went with a pleasant walk for one participant ($+0.78$) and with an unpleasant one for another (-0.78).

What these measurements open up?

From ratings to signatures. Each evaluated term has a recorded counterpart: effort in leg EMG, stability in the margin of stability, control and predictability in EEG agency and prediction-error signatures [7].

Weights per person. The block-to-block comparisons allow the weights w_i to be estimated for each wearer [6], as a candidate objective for tuning assistance beyond metabolic cost.

$$\text{Judgment}(\text{mode}) = \sum w_i \cdot \text{term}_i \quad w_i = ?$$

The judgment rated after every block:
pleasant or unpleasant | easier or harder than no device | better or worst than the last block | accept it for 30 more minutes?

CANDIDATE TERM	EVALUATED (QUESTIONNAIRE, EVERY BLOCK)	RECORDED (BODY AND BRAIN SIGNALS)
$w_1 \cdot \text{Effort}$	Borg CR10, legs and breathing [4]	Net metabolic energy (day 1), leg EMG (both days)
$w_2 \cdot \text{Stability}$	“I felt steady on my feet.”	margin of stability [5], step-width variability
$w_3 \cdot \text{Control}$	agency x3, predictability, trust and safety x2	EEG: candidate agency and prediction-error signatures
$w_4 \cdot \text{Naturalness}$	“The walking felt natural.”	deviation from no-exo gait (IMU, GRF)
$w_5 \cdot \text{Attention}$	“I had to concentrate on my walking.”	EEG (candidate markers)
$w_6 \cdot \text{Comfort}$	Discomfort by body region, restrictions, embodiment x3	self-report only
+ ...	spoken or written: “If you could change one thing ...”	terms we did not anticipate

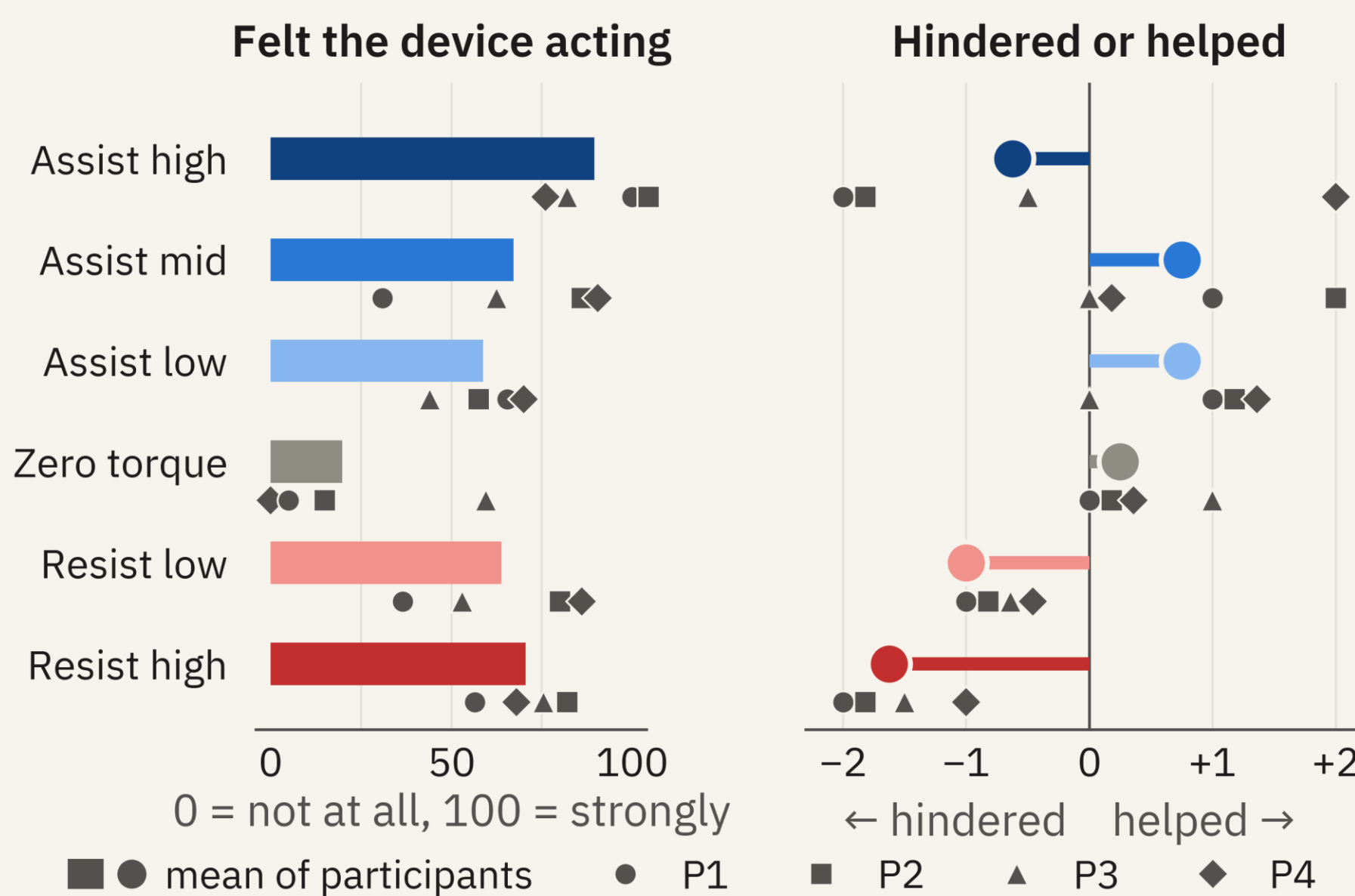


Evaluation

First-pass analyses: 4 participants, 8 blocks per day, 2 days; calorimetry on day 1.

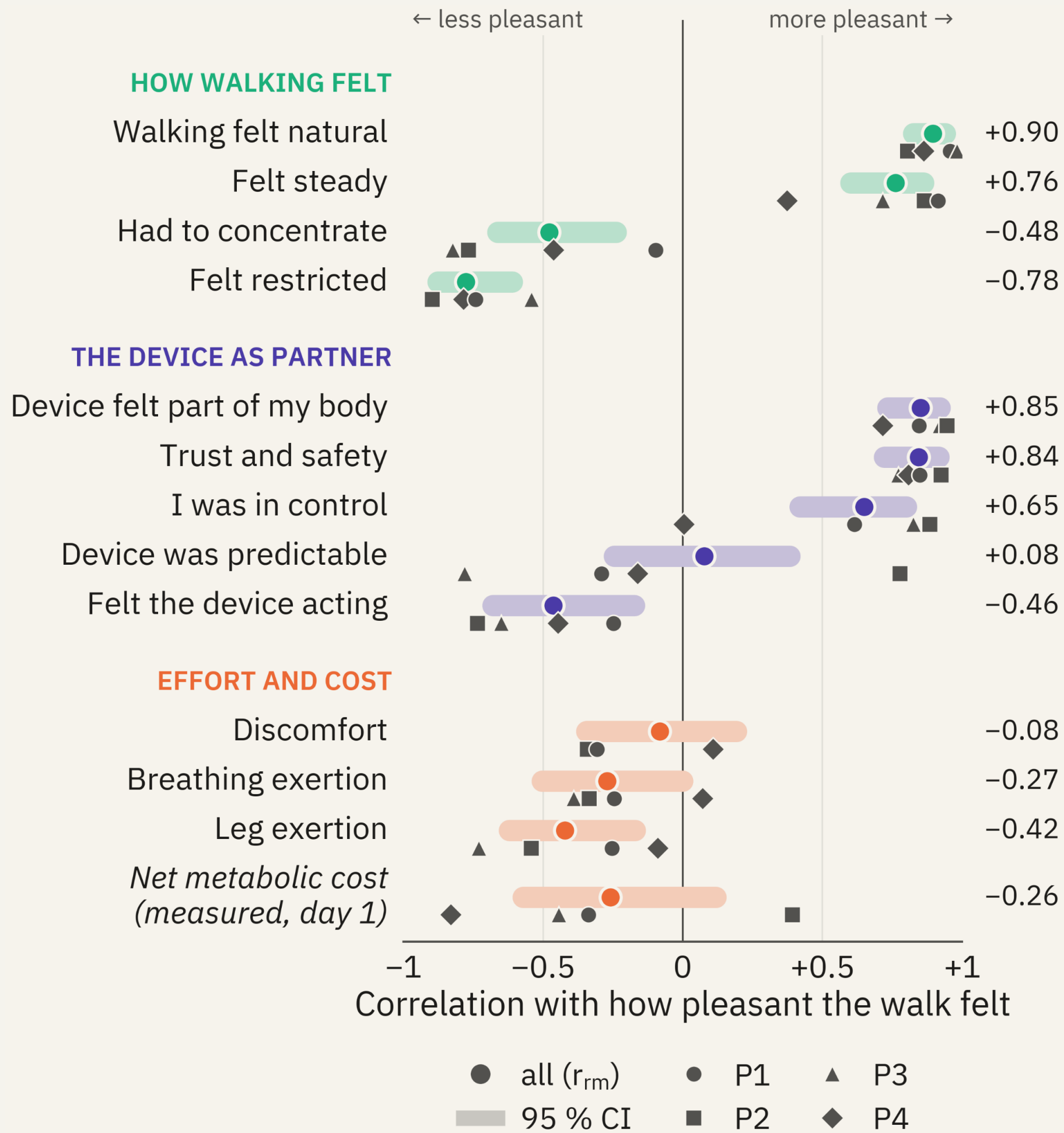
A Do participants feel the direction?

Rated after every block; each participant's values are averaged over the days they rated. Resistive modes hindered, low and mid assistance helped. High assistance, the mode felt most strongly, was rated hindering by three of four participants.



B What goes with a pleasant walk?

Within-participant correlation [6] of each term with how pleasant the walk felt, rated after every block.

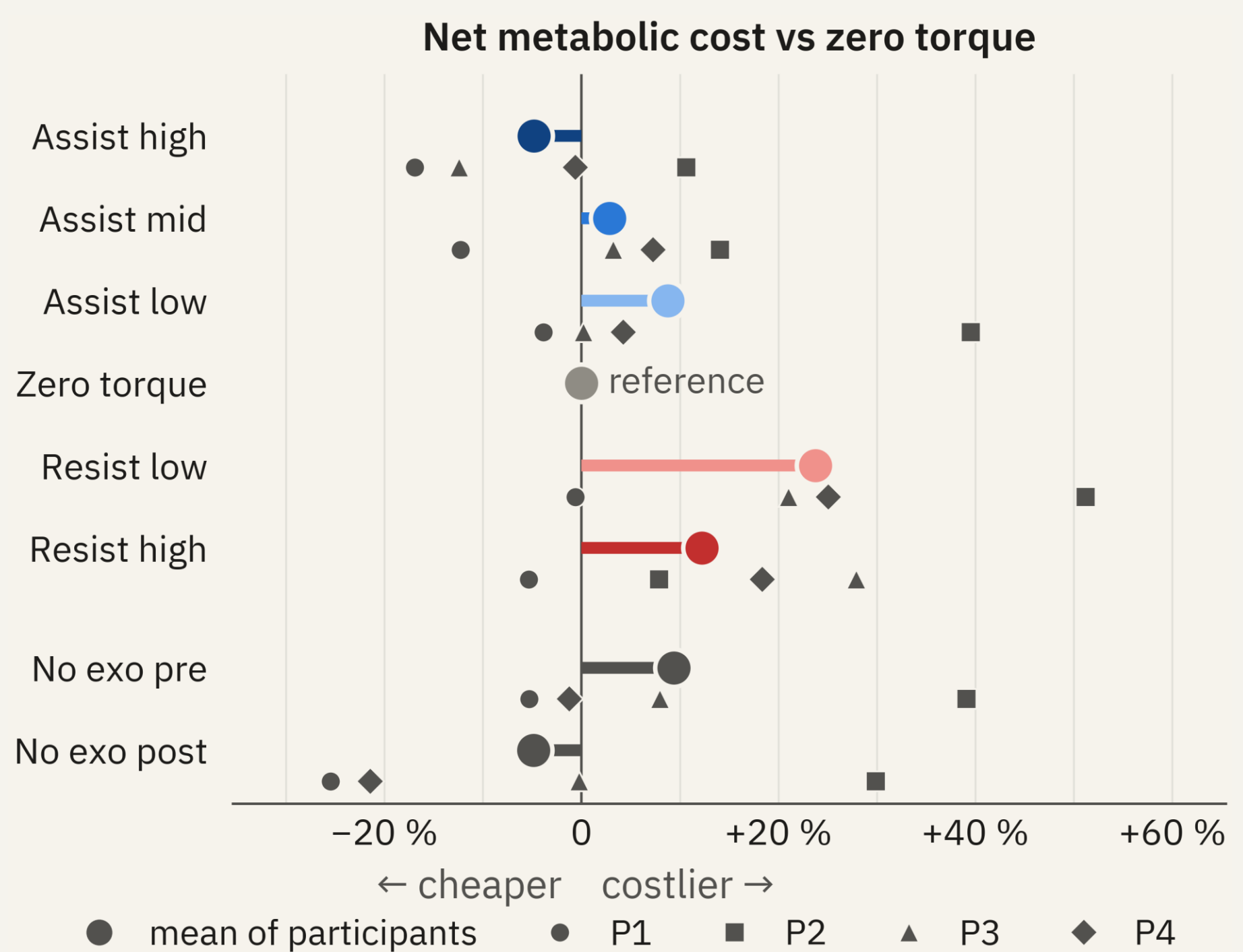


Body & Brain

The recorded counterparts of the evaluated terms, from the same walking blocks; calorimetry on day 1.

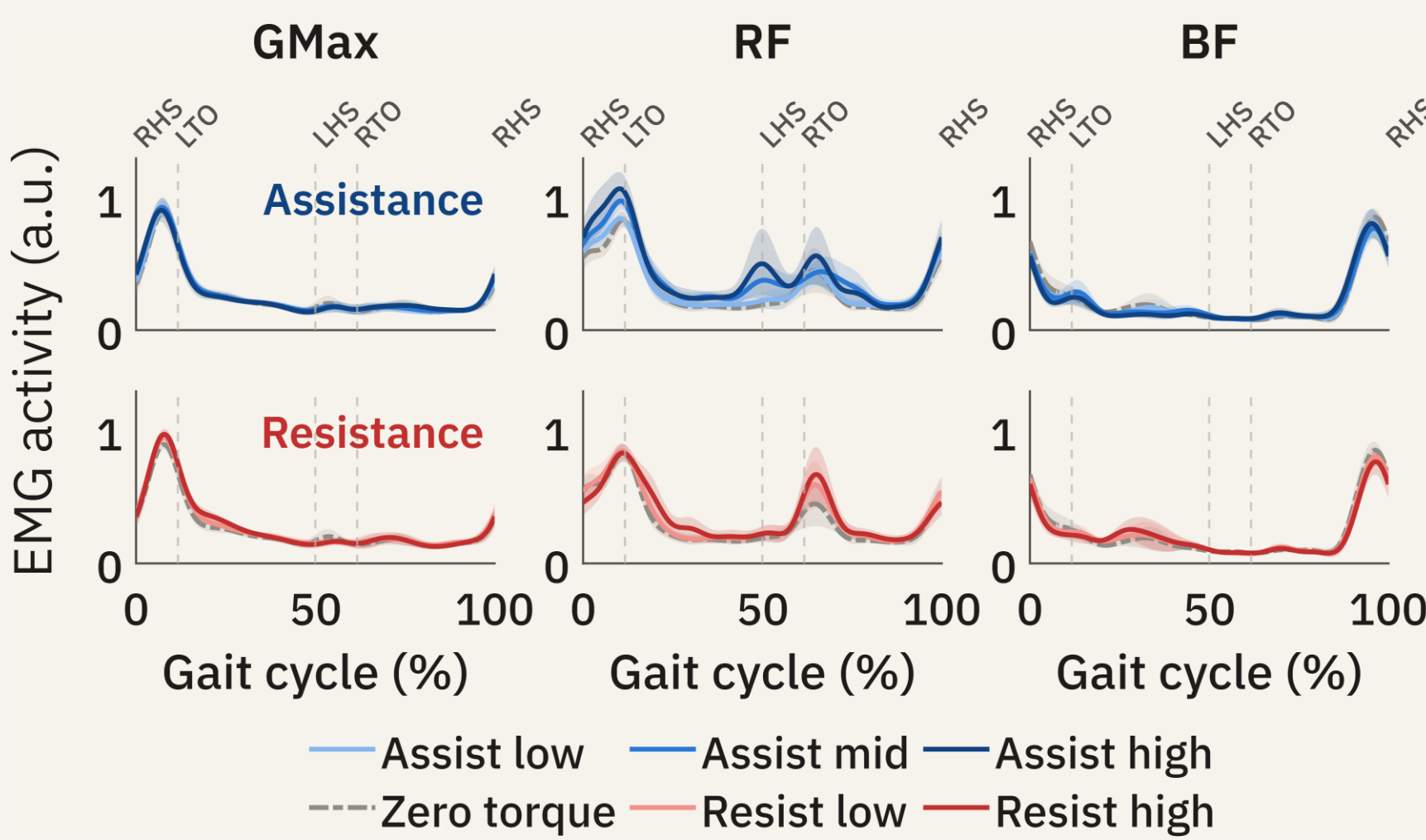
c What goes with a pleasant walk?

Net cost of transport, measured on day 1. Resistive modes cost about 10 to 25 % more and high assistance about 5 % less. Low and mid assistance, rated as helping (A), did not lower the cost. Without the exo, cost was 9 % higher at the start of the session and 5 % lower at the end.



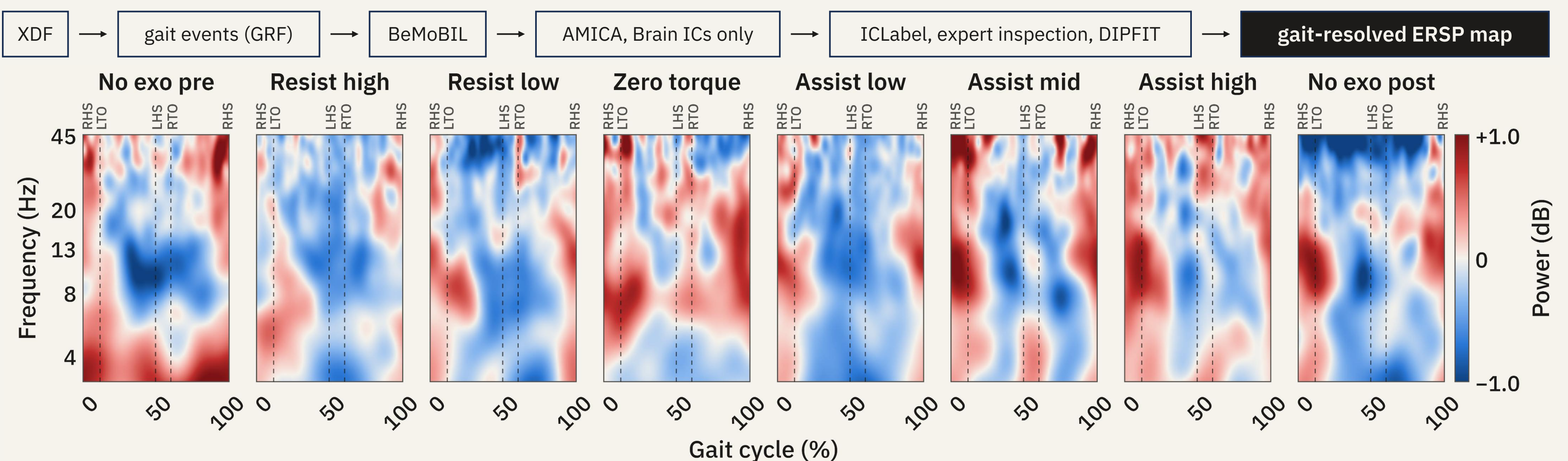
d How do the muscles respond to each mode?

Mean $\pm$ SEM across participants, both legs; 1 = median zero-torque peak. RF activity rose around toe-off with high assistance and high resistance; GMax and BF changed little.



e What happens in the brain during exo walking?

Right primary motor cortex Event Related Spectral Perturbation (4 participants) In most blocks, alpha and beta power (8–30 Hz) is higher around right heel strike. Changing the mode shifted power consistently across participants.



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